Basics of Operating System

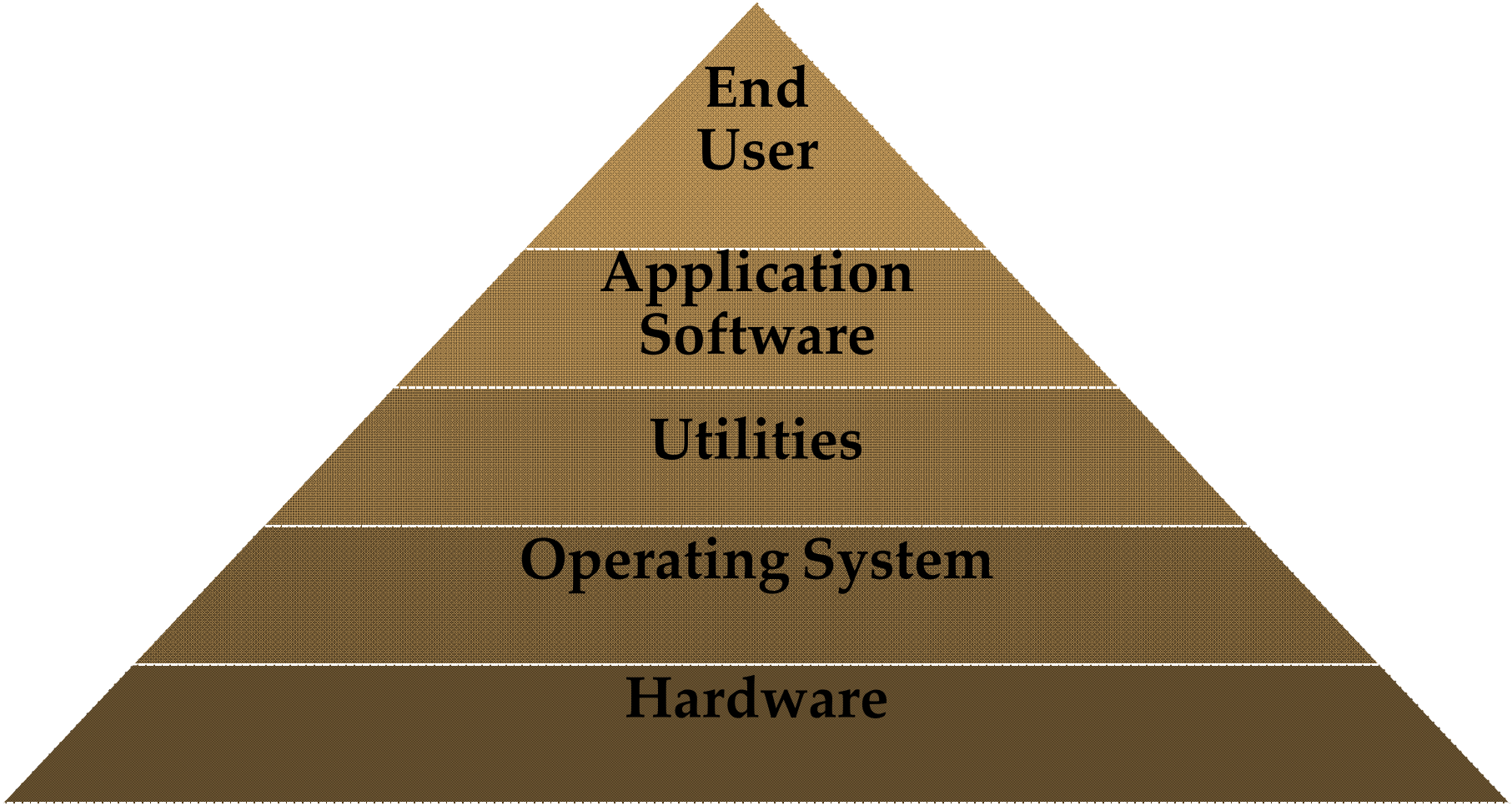
SWE 4101
Introduction to Software Engineering

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Operating system

- Operating system (OS) is system software that controls and coordinates the use of hardware among the different application software and users. OS intermediates between the user of computer and the computer hardware.

Components of Computer System



Objectives of OS

- The user gives a command and the OS translates the command into a form that the machine can understand and execute.
- OS has two main objectives—
  - (1) to make the computer system convenient and easy to use, for the user, and
  - (2) to use the computer hardware in an efficient way, by handling the details of the operations of the hardware

Types of OS

- OS are classified into different types depending on their capability of processing—

1) Single user,
2) Multiuser,
3) Multitasking,
4) Multiprocessing,
5) Real time, and
6) Embedded.

Function of OS

- Process Management
- Memory Management
- File Management
- Device Management
- Protection and Security
- User Interface and Command Interpreter

Multiprogramming

- Uniprogramming:
  - Only one process is at a time in execution
  - All system resources are exclusively available

- Multiprogramming:
  - Interleaved execution of two or more different and independent programs by a computer.
  - Area occupied by each job residing simultaneously in the main memory is known as a memory partition.

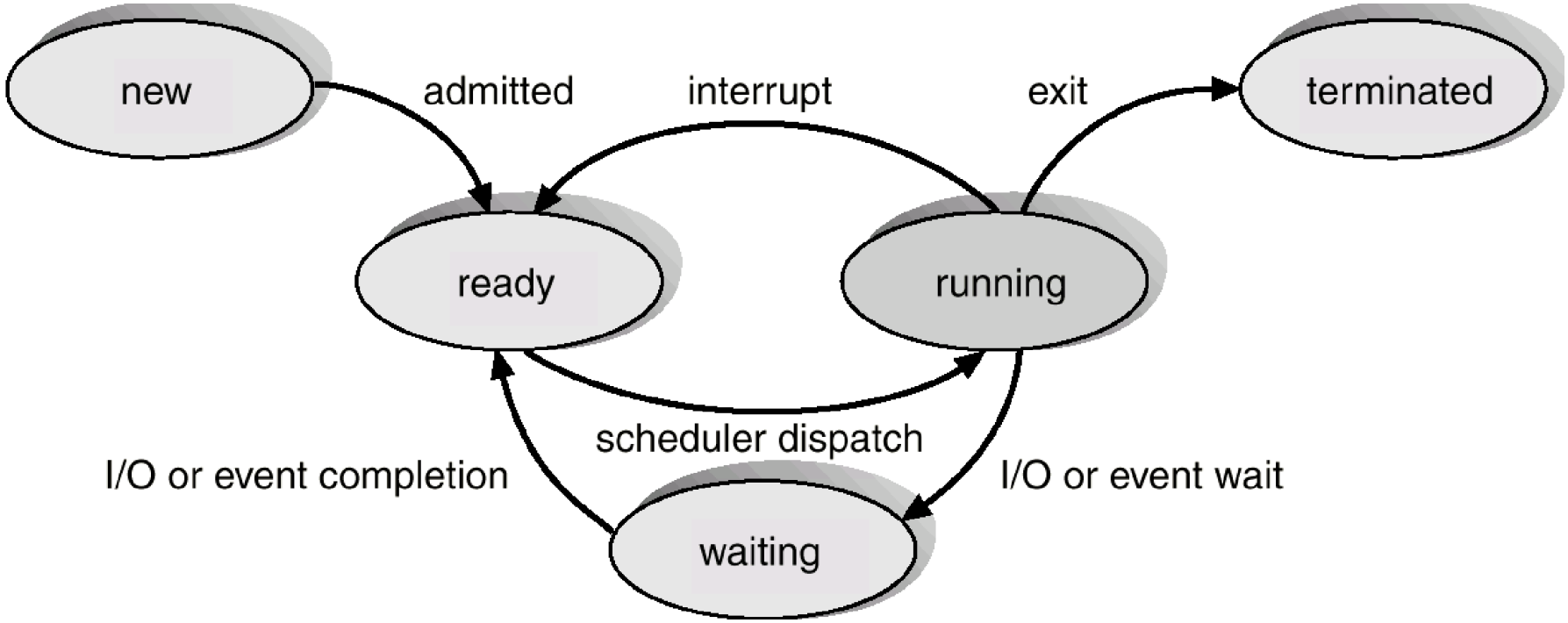
Process Management

- A process is a program in a state of execution.
- It is a unit of work for the operating system.
- A process can be created, executed, and stopped.
- A program is always static and does not have any state.
- A program may have one or more processes running.

Process State

- New—process is in a new state when it is created,
- Ready—process is in ready state when it is waiting for a processor,
- Running—process is in running state if processor is executing the process,
- Waiting—process is in waiting state when it waits for some event to happen (I/O etc), and
- Terminated—process that has finished execution is in terminated state.

Process State



CPU Scheduling

- Different types of CPU Scheduling Algorithms:
  - First Come First Serve (FCFS)
  - Shortest Job First (SJF)
  - Round Robin (RR)

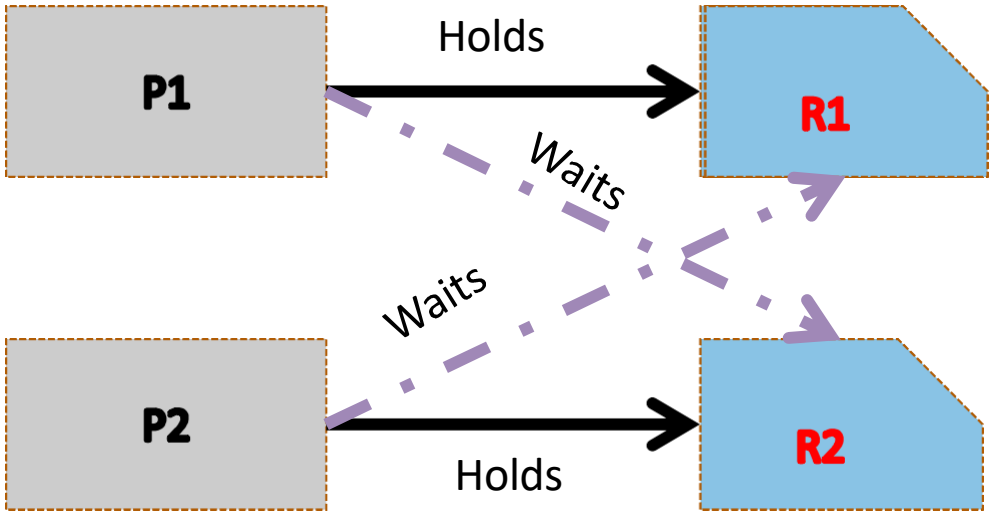
- Pre-emptive:
  - CPU can withdraw any process from execution.

Process Synchronization

- Concurrent Processes
  - Two or more process independent of each other execute at the same time.
- Race Condition
  - Multiple process accessing and manipulating same data and the final result depends on the order of execution.

Deadlock

- A process waits endlessly for a resource and the requested resource is being used by another process that is waiting for some other resource



Deadlock

- Four necessary Conditions:
  - Mutual Exclusion
  - Non pre-emption
  - Holds and wait
  - Circular wait
- Handling
  - Deadlock Prevention
  - Deadlock Avoidance

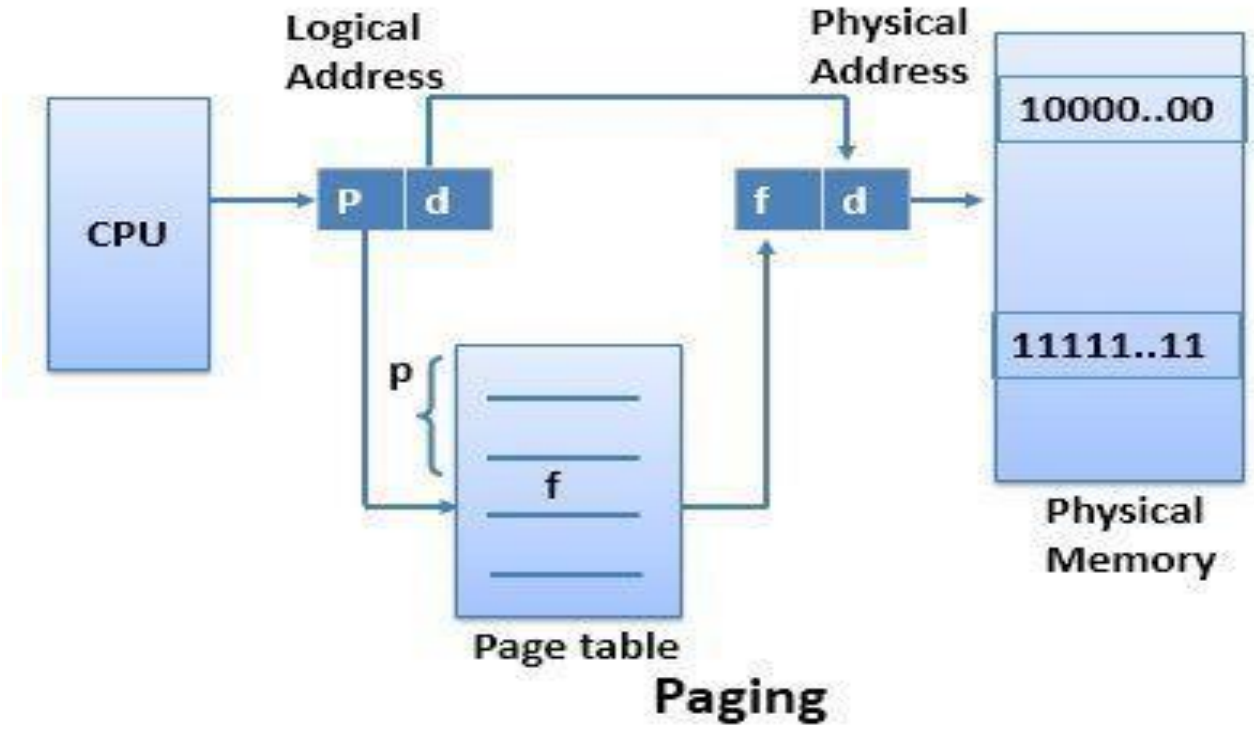
Memory Management

- In multiprogramming environment memory management is crucial.
- The single block of available memory is called a hole.

- Hole Allocation Strategy
  - First fit
  - Best fit
  - Worst fit
- ## Causes fragmentation

Memory Management

- Paging



Paging

File Management

- A set of files and
- A directory structure

Device Management

- OS manages and controls the devices attached to the computer.
- The OS communicates with the I/O hardware via the device driver software. The device driver software comes along with each device

Protection And Security

- The access of programs, processes, and users, to the resources defined by the computer are controlled by the protection mechanism.
- Security can be ensured using—
- (1) user accounts—individual accounts for each user,
- (2) user authentication—using password protection,
- (3) access rights—define rights for access of different kind of information for different users,
- (4) data encryption—store data in computer in encrypted form, and
- (5) data backup—storing data on a peripheral device other than the hard disk.

User Interface

- Make the computer convenient for use by its user
- Two kinds of interfaces—
  - (1)Command Line Interface (CLI), and
  - (2) Graphical User Interface (GUI).

Reference

- Follow the book Computer Fundamentals by Anita Goel (chapter 7)